

UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

DEPARTMENT OF COMPUTER SCIENCE

Object-Oriented Programming

Homework 4-Project Definition

Group N°1

Project: Liquor Store System

Team Name: Logical Lords

Members:

Isaac Sebastian Acurio Catota

Armas Guaño Demian Rolando

Enrique Mateo Ayala Lagunas

Instructor: Edison Lascano PhD

NRC: 36862

October 7, 2026

Quito – Ecuador

Problem

The liquor store currently manages its operations through a combination of manual records and information kept on a mobile phone. These records are not centralized, making it difficult for the owner to maintain a consistent view of product quantities, customer debts, and cash movements.

Inventory management is one of the main concerns because the owner needs to know which products are available and which require restocking. Supplier information and pending purchase orders also need organized records to support replenishment. Similarly, customer credit purchases and subsequent payments must be recorded consistently so that outstanding balances can be checked accurately.

At the end of the day, the owner needs to compare the expected cash balance with the money physically available. This requires distinguishing between sales, money actually received, customer debt payments, and expenses. The owner also needs to identify which sales were made personally and which were made by the employee.

The business requires a clear pricing rule to calculate selling prices and evaluate the expected gross profit from product sales. Discounts must also follow defined conditions to prevent inconsistent charges and calculation errors.

These difficulties create a risk of stock shortages, incorrect customer balances, pricing errors, and unexplained cash differences. The project therefore addresses the need for a centralized system that records these operations and provides reliable information for daily business decisions.

Overview

Liquor Store System arises from the need to overcome the manual limitations and disorganization affecting the liquor store’s daily operations. Its main objective is to centralize business information and help the owner maintain accurate inventory records, organize customer debts, monitor sales, and reconcile cash movements.

The system will provide an updated view of available inventory after each recorded operation, helping the owner identify products that require restocking. It will also organize supplier information and pending purchase orders, supporting the replenishment process. A dedicated section for credit sales will record customer debts and payments, allowing the owner to check who owes money and each customer’s outstanding balance.

To support daily cash reconciliation, the system will distinguish between sales made by the owner and those made by the employee. It will record payments received at the time of sale, collections of outstanding customer debts, and expenses so that the expected cash balance can be compared with the amount physically available. The system will also support the calculation of selling prices and gross profit according to agreed business rules, automatically apply discounts when their conditions are met, and provide reports on sales and best-selling products. Before completing a sale, the system will remind the cashier to verify the customer's age and request confirmation that the check has been performed.

The project will be developed in stages, initially focusing on the operations of a single liquor store. This approach will allow the team to validate its behavior with the client and extend it as needs are confirmed. The expected result is more consistent records, fewer calculation errors, and reliable information for purchasing and daily management decisions.

Background

The project is based on the daily operations of a liquor store managed by its owner with the support of an employee. The business currently combines manual records with information kept on a mobile phone to manage activities such as sales, inventory, customer credit, and payments.

These activities are closely related. Completed sales reduce the quantities of available products, while supplier deliveries replenish inventory. Purchase orders help organize restocking, and information about best-selling products supports purchasing decisions. Purchase costs, selling prices, and discounts also influence the gross profit obtained from sales.

Customer credit introduces an additional distinction between selling a product and receiving payment. A credit sale creates an outstanding balance, while subsequent payments reduce that debt and must be recorded separately. Similarly, daily cash reconciliation requires identifying the money received, expenses paid, and the person responsible for each operation.

This operational context motivates the development of Liquor Store System as an academic project connected to a real client. Understanding these relationships provides the basis for defining the system’s requirements and developing a solution that can be validated with the owner and expanded in later stages.